PART 1 - GENERAL

1.1 Summary

23 81 23 Part 1.1.1.A This section specifies the requirements for chilled water Computer Room Air Handling (CRAH) units for the Meridian Data Centre Campus, Phase 1. CRAH units shall provide precision sensible cooling to each Server Hall using the campus chilled water plant.

1.2 References

23 81 23 Part 1.2.1.A The following standards shall apply: AHRI 1360 (Performance Rating of Computer and Data Processing Room Air Conditioners), ASHRAE 90.1 (Energy Standard for Buildings), IS 1391 (Room Air Conditioners), ASHRAE TC 9.9 (Thermal Guidelines for Data Processing Environments).

1.3 Submittal Requirements

23 81 23 Part 1.3.1.A Submittals shall include certified cooling capacity data, airflow performance curves, chilled water coil ratings, fan performance data, dimensional drawings, electrical data, and BMS integration details.

1.4 Quality Assurance

23 81 23 Part 1.4.1.A CRAH manufacturer shall have a minimum of 10 years experience in the manufacture of precision cooling systems for mission-critical data centre environments. Manufacturer shall hold ISO 9001 certification.

PART 2 - PRODUCTS

2.1 Acceptable Manufacturers

23 81 23 Part 2.1.1.A Acceptable CRAH manufacturers: CryoCore Climate, or approved equivalent meeting all performance requirements of this section.

2.2 Performance and Design Criteria

23 81 23 Part 2.2.1.A Each CRAH unit shall provide a minimum sensible cooling capacity of 250 kW at the design conditions specified herein.

23 81 23 Part 2.2.1.B Design entering chilled water temperature shall be 10°C. The CRAH cooling coil shall be selected and rated at this entering water temperature.

23 81 23 Part 2.2.1.C Design leaving chilled water temperature shall be 18°C, resulting in a chilled water temperature differential (delta-T) of 8 K across the cooling coil.

23 81 23 Part 2.2.2.A Minimum unit airflow rate shall be 64,320 m^3/h at the design external static pressure. Airflow shall be measured at the unit discharge per AHRI 1360 test conditions.

23 81 23 Part 2.2.3.A Supply air temperature shall be 23°C +/- 1°C at design conditions. Return air temperature shall be 35°C maximum under hot aisle containment.

23 81 23 Part 2.2.4.A Total cooling capacity (sensible plus latent) shall be submitted along with the sensible heat ratio (SHR) at design conditions.

23 81 23 Part 2.2.4.B Cooling capacity shall be verified by the manufacturer at the specified entering and leaving water temperatures and air conditions.

23 81 23 Part 2.2.4.C Unit coefficient of performance (COP) shall be a minimum of 5.2. The vendor shall submit the COP value as part of the performance data submittal.

23 81 23 Part 2.2.5 CRAH unit shall maintain operation across a relative humidity range of 20% to 80% non-condensing within the server room environment. Humidity control shall be integrated with the unit controller.

23 81 23 Part 2.2.6.A CRAH unit controller shall provide proportional-integral (PI) control of supply air temperature with adjustable setpoints. Controller shall support Modbus TCP/IP and BACnet communication protocols.

23 81 23 Part 2.2.7.A Unit shall include high-efficiency air filters with a minimum MERV 11 rating. Filter frames shall be accessible for replacement without tools.

23 81 23 Part 2.2.8 Where the CRAH unit incorporates a direct expansion (DX) trim cooling circuit, the refrigerant type shall be R-410A only. Use of R-407C, R-22, or any other refrigerant is not permitted.

2.3 Component Description

23 81 23 Part 2.3.1.A Chilled water cooling coil shall be copper tube with aluminium or copper fins. Coil shall be leak tested at the factory at 1.5 times working pressure. Coil connections shall be flanged.

23 81 23 Part 2.3.2.A CRAH unit fans shall be electronically commutated (EC) type with variable speed drive. Fan motor power density shall be less than 0.10 W per m^3/h of airflow at design conditions.

23 81 23 Part 2.3.2.B Fan motor power shall be specified in electrical kilowatts (kW). Specifications of motor power in alternative units (HP, BHP, or PS) shall not be accepted.

23 81 23 Part 2.3.3.A Fan assemblies shall be direct-drive with integrated EC motors. Belt-driven fan arrangements are not acceptable.

23 81 23 Part 2.3.4.A Unit shall include a condensate drain pan of stainless steel construction with dual drain connections. Drain pan shall be insulated to prevent external condensation.

23 81 23 Part 2.3.5.A CRAH unit shall be equipped with a 2-way modulating control valve for chilled water flow regulation. The valve flow coefficient (Cv) shall be specified and submitted for hydraulic circuit balancing calculations.

23 81 23 Part 2.3.5.B Control valve actuator shall be spring-return, fail-closed type with a modulating signal range of 0-10 VDC or 4-20 mA.

23 81 23 Part 2.3.5.C Chilled water flow regulation shall utilise pressure independent control valves (PICV) to ensure consistent flow regulation regardless of system pressure variations. Standard 2-way modulating valves without pressure independence are not acceptable.

23 81 23 Part 2.3.6.A Unit shall include onboard humidity sensing with dry bulb and dew point temperature sensors.

23 81 23 Part 2.3.6.B Humidity control shall be provided by an electrode steam generating humidifier integrated within the CRAH unit. Infrared, ultrasonic, or spray-type humidifier systems are not acceptable.

2.4 Physical Construction

23 81 23 Part 2.4.1.A CRAH unit overall cabinet width shall not exceed 2400 mm to ensure transport clearance through standard data centre corridors and freight elevator openings. Units exceeding this width shall not be accepted.

23 81 23 Part 2.4.1.B Cabinet construction shall be double-skin insulated panels with powder coated galvanised steel exterior finish. Internal lining shall be acoustically treated.

23 81 23 Part 2.4.2.A Unit shall be suitable for underfloor air discharge configuration with bottom discharge plenum. Side return air inlet arrangement shall be provided.

PART 3 - EXECUTION

3.1 Examination and Preparation

23 81 23 Part 3.1.1.A Verify that chilled water piping, floor grilles, and electrical connections are complete and ready prior to CRAH unit placement.

3.2 Installation

23 81 23 Part 3.2.1.A CRAH units shall be placed on anti-vibration pads on the prepared raised floor structure. Chilled water connections shall be made with flexible connectors and isolation valves.

3.3 Field Testing and Commissioning

23 81 23 Part 3.3.1.A Perform chilled water system flushing and water quality verification prior to CRAH unit startup.

23 81 23 Part 3.3.2.A Verify sensible cooling capacity is a minimum of 250 kW at the specified entering chilled water temperature. Measurements shall use calibrated temperature sensors and flow meters.

23 81 23 Part 3.3.2.B Verify chilled water flow rate through the CRAH cooling coil matches the design value within +/- 10%. Flow shall be measured using an ultrasonic flow meter.

23 81 23 Part 3.3.2.C Verify entering and leaving chilled water temperatures match the design basis values. Temperature measurements shall use calibrated RTD sensors with accuracy of +/- 0.1°C.

23 81 23 Part 3.3.2.D Verify supply air temperature is within the specified tolerance of 23°C +/- 1°C at design load conditions.

23 81 23 Part 3.3.2.E Verify unit airflow rate at design external static pressure using a calibrated airflow measurement station or traverse method per ASHRAE 111.

23 81 23 Part 3.3.2.F Verify EC fan motor current draw is within the nameplate rating. Fan speed shall be set to achieve the specified airflow rate.

23 81 23 Part 3.3.2.G Verify humidity control operation. Humidifier shall maintain room humidity within the specified range during the test period.

23 81 23 Part 3.3.2.H Verify BMS communication and alarm integration. All critical alarms (high supply temperature, low airflow, chilled water leak, filter differential pressure) shall be confirmed operational.

23 81 23 Part 3.3.2.I Verify condensate drain system operation under full cooling load. Drain pan shall show no evidence of overflow or leakage during the test.

23 81 23 Part 3.3.2.J Perform 72-hour continuous run test under simulated IT load. CRAH unit shall maintain stable supply air temperature throughout the test period without nuisance alarms or shutdowns.